

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Civil Engineering End Term Examination, Academic Year: 2025-26 (Sem – I)		
	Class: Second Year B.Tech.	Course: Building Construction	Course Code: PCC-CE204
	Date: 18.11.2025	Time: 13:30 to 15:30	Max. Marks: 60

Q.N.	Questions	Marks	CO	BT
1.	Explain the different types of shallow and deep foundations with neat sketches? State the selection criteria of foundations.	10	2	2
2.	Enlist and explain the different types of floors in detail.	10	1	2
3.	Explain in detail with sketches the different types of stairs. Give the requirements of good stairs.	10	2	2
4.	Discuss the different types of slip formwork. Give its advantages and disadvantages.	10	5	1
5.	Give the detailed classification of building materials.	10	3	2
6.	What are the causes and effects of dampness in structures? Explain the various methods of damp proofing.	10	4	3

Bloom's Taxonomy (BT)

1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyze, 5 - Evaluate, 6 – Create

Course Outcomes (COs):

1	Enlist the components of a building.
2	Understand the role and importance of various building components and its uses.
3	Apply properties of various construction materials as per the requirement.
4	Analyses and identify the quality materials for construction activity.
5	Evaluate performance of a building with inclusion of various building services.



S. G. G. S. Institute of Engineering and Technology
Vishnupuri, Nanded-431606, India
Department of Civil Engineering

END SEMESTER EXAMINATION – 2025-26 (Semester-I)

Date: Wednesday, 19 November 2025 Course: Surveying and Geomatics
Class: Second Year Civil Engineering Course Code: PCC-CE203

Time: 13:30 – 15:30
Maximum Marks: 60

- Note
1. Attempt all questions.
 2. Use of non-programmable calculator is permitted.
 3. Draw appropriate figures wherever required.
 4. Assume appropriate data, if necessary.

Que. No.	Question statement	Mark	CO	BT
01	Calculate the reduced bearing based on following readings, a) $359^{\circ}20'20''$, b) $245^{\circ}15'00''$, c) $135^{\circ}00'00''$	06	CO2	BT3
02	Define contour and write characteristics of contours.	06	CO5	BT3
03	Explain the method of intersection of plane table surveying with neat diagrams.	06	CO1	BT2
04	Two straight roads meet at an angle of 130° . Determine the necessary data for setting out a circular curve of 300 m radius between the road by perpendicular offset method.	06	CO4	BT4
05	A road embankment is 30 m wide at the top with side slopes of 2:1. The ground level at two points A and B is 170.3 and 169.1 m. The distance between point P and Q is 100 m. The formation level at A is 178.7 m with a uniform falling gradient of 1 in 50 from A to B. Determine the volume of earthwork between A and B using trapezoidal formula.	06	CO3	BT4
06	Discuss segments of GPS.	06	CO5	BT2
07	Write a short note on 'remote sensing and its application as modern tool for surveying'.	08	CO5	BT3
08	Determine the level for the levelling work performed using an auto level for centreline of canal distributary. The levelling work was started from the soffit of the bridge over a river. The level of soffit of bridge girder was 395 m. Subsequent reading were taken at interval of 20 m as written below. The instrument was shifted after 5 th , 7 th , and 9 th reading. Use HI method. -1.025, 0.255, 0.155, 0.055, 0.020, 1.545, 0.065, 1.025, 0.865, 1.275, 0.865, 1.025, 0.155	08	CO2	BT4
09	Determine the horizontal distance between point P and Q. The Tacheometer is on point P and height of the instrument from ground is 1.480 m. The vertical angle at P is $+5^{\circ}20'$. The levelling staff is kept vertical at Q. The observed staff readings were 0.545 m, 0.905 m, and 1.265 m. Assume $K = 100$ and $s = 0$. Also, determine the elevation of Q if the elevation of P is 200 m.	08	CO3	BT4

* * * All the Very Best * * *

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1 Define principles of Surveying, Remote Sensing and Geomatics	BT1 - Remember
CO2 Describe different instruments, tools, applications and techniques to determine the positions on the surface of the earth, change detection	BT2 - Understand
CO3 Apply the concepts of modern surveying techniques and Instrumentation.	BT3 - Apply
CO4 Differentiate the techniques for setting out alignments, curves, other layouts, modern survey systems etc.	BT4 - Analyse
CO5 Formulate basic and modern Surveying, Remote Sensing and Geomatics techniques to be used for a specific civil engineering project	BT5 - Evaluate
	BT6 - Create

*** All the Very Best ***



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Civil Engineering End Sem. Examination Academic Year: 2025-26 (Semester I)

Class: S.Y. B. Tech. (Civil Engg.)

Course: Fluid Mechanics

Course Codes: CW202, PCC- CE202 & PCC-CE203

Date: 24/11/2025

Time: 01:30 P.M. - 03:30 P.M

Max. Marks: 60

Note: i. Assume suitable data whenever necessary and state it clearly. ii. Mobile phones and programmable calculators are strictly prohibited. iii. Exchange/Sharing of stationery, calculator etc. not allowed. iv. Figures to right indicate full marks. v. The symbols carry usual meaning.

Q. No.	Question	Marks	CO	BT
Q.1	Explain in brief about the following physical properties of fluid: (a) Specific gravity (b) Surface tension. A clean tube of internal diameter 3 mm is immersed in a mercury with a coefficient of surface tension as 0.48 N/m. The angle of contact of the mercury with the glass can be assumed to be 120° . Take the specific gravity of mercury as 13.6. Determine the amount of capillary depression.	12	CO1	BT1 & BT3
Q.2	Explain in brief about (i) Pressure prism (ii) Centre of pressure. A vertical isosceles triangular gate with its vertex up has a base width of 2 m and a height of 1.5 m. If the vertex of the gate is 1 m below the free water surface, find the total pressure force and the position of the centre of pressure.	12	CO2	BT2 & BT3
Q.3	Explain in brief about velocity potential and stream function. For the velocity potential function, $\phi = 5(x^2 - y^2)$ calculate the velocity components at a point (2, 3).	12	CO3	BT4
Q.4	What is coefficient of discharge for flow through a rectangular notch? Obtain the relationship between shear stress and pressure gradient for laminar flow through circular pipe.	12	CO4	BT5
Q.5	Give the expressions for major head loss and loss of head due to sudden contraction for a flow through pipe. A syphon of diameter 200 mm connects two reservoirs having a difference in elevation of 20 m. The length of syphon is 500 m and the summit is 3 m above the water level in the upper reservoir. Determine the discharge through the syphon.	12	CO5	BT2 & BT3